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Chapter 11: Simple Linear Regression and Correlation

Learning objectives

1. Empirical Models
2. Simple Linear Regression
3. Correlation

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- Many problems in engineering and science involve exploring the relationships between two or more variables.
- **Regression analysis** is a statistical technique that is very useful for these types of problems.
- For example, in a chemical process, suppose that the yield of the product is related to the process-operating temperature.
- Regression analysis can be used to build a model to predict yield at a given temperature level.

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Table 11-1 Oxygen and Hydrocarbon Levels

Observation Number	Hydrocarbon Level x (%)	Purity y (%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33

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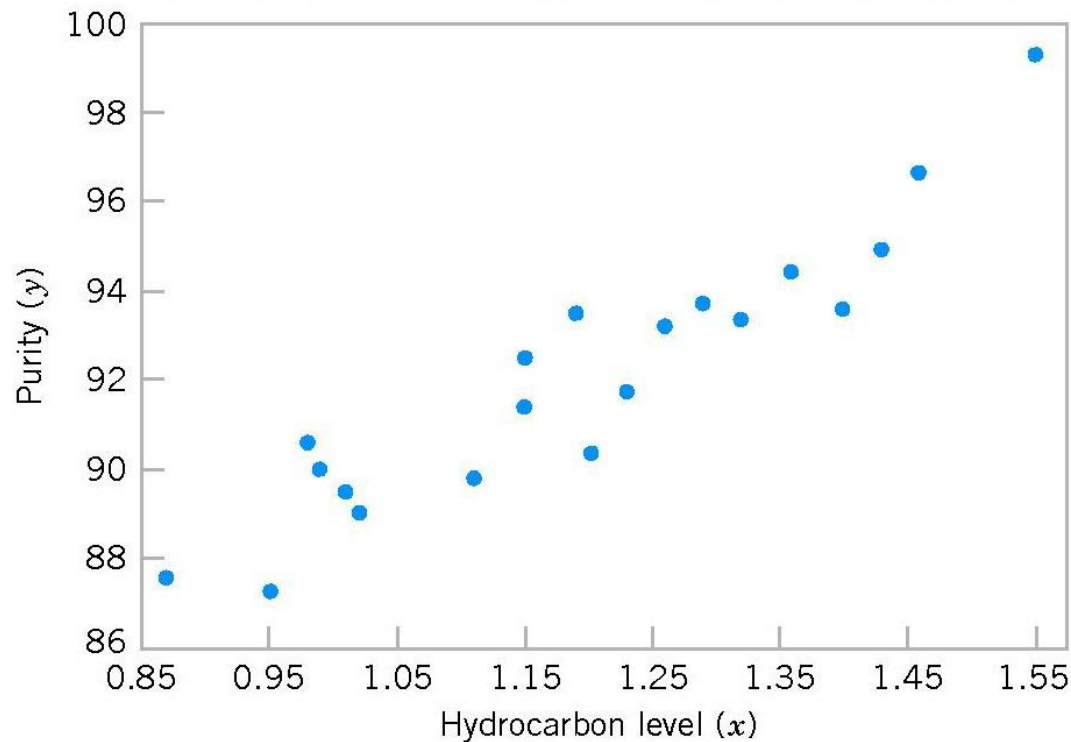


Figure 11-1 Scatter Diagram of oxygen purity versus hydrocarbon level from Table 11-1.

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Based on the scatter diagram, it is probably reasonable to assume that the mean of the random variable Y is related to x by the following straight-line relationship:

$$E(Y | x) = \mu_{Y|x} = \beta_0 + \beta_1 x$$

regression coefficients.

The **simple linear regression model** is given by

$$Y = \beta_0 + \beta_1 x + \varepsilon$$

random error

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Suppose that the mean and variance of ε are 0 and σ^2 , respectively, then

$$E(Y | x) = E(\beta_0 + \beta_1 x + \varepsilon) = \beta_0 + \beta_1 x + E(\varepsilon) = \beta_0 + \beta_1 x$$

The variance of Y given x is

$$V(Y | x) = V(\beta_0 + \beta_1 x + \varepsilon) = V(\beta_0 + \beta_1 x) + V(\varepsilon) = 0 + \sigma^2 = \sigma^2$$

The true regression model is a line of mean values:

$$\mu_{Y|x} = \beta_0 + \beta_1 x$$

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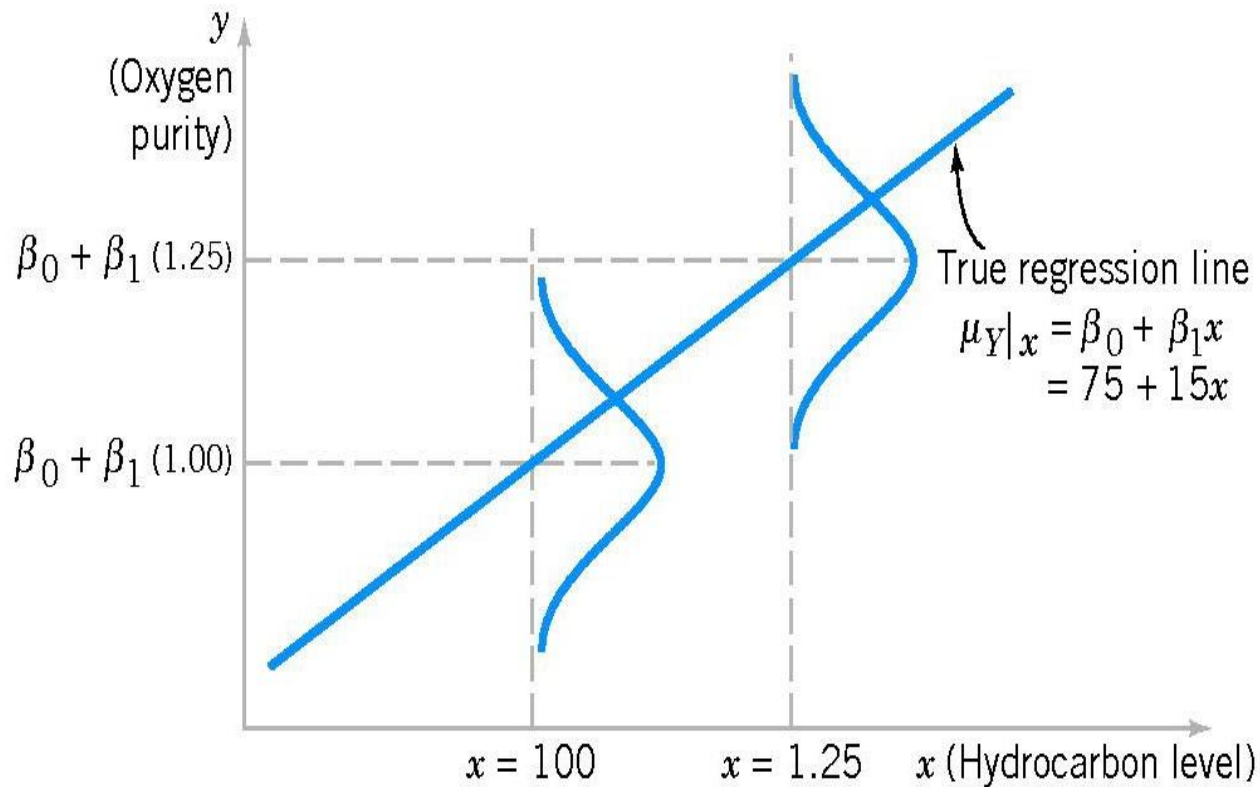


Figure 11-2 The distribution of Y for a given value of x for the oxygen purity-hydrocarbon data.

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- The case of simple linear regression considers a single regressor or **predictor** x and a dependent or **response variable** Y .

- The expected value of Y at each level of x is a random variable:

$$E(Y | x) = \beta_0 + \beta_1 x$$

- We assume that each observation, Y , can be described by the model

$$Y = \beta_0 + \beta_1 x + \varepsilon$$

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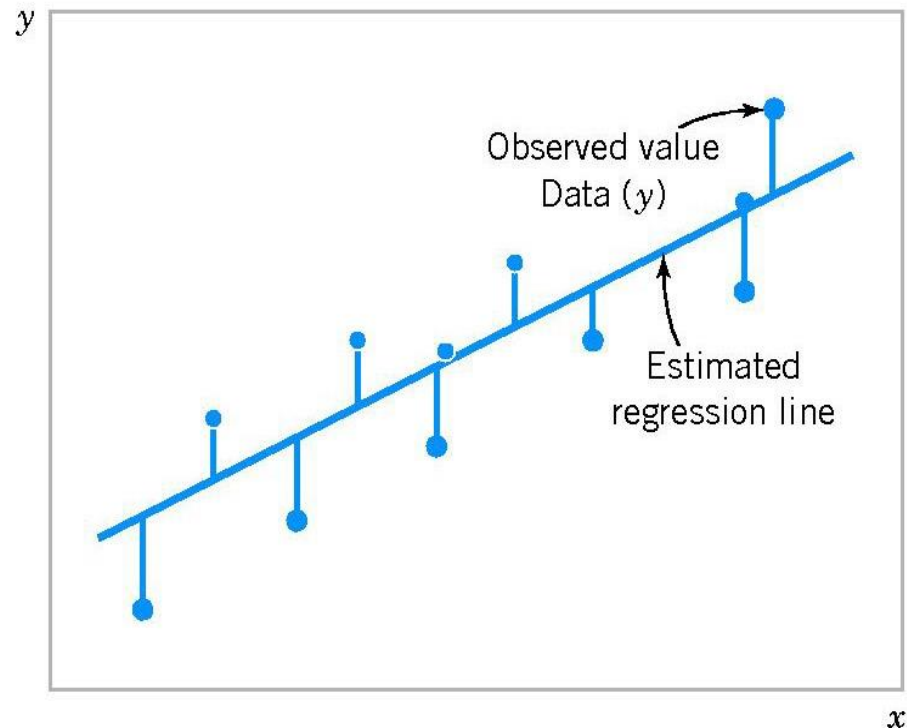
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Suppose that we have n pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i=1, \dots, n$$

Figure 11-3
Deviations of the data from the estimated regression model.



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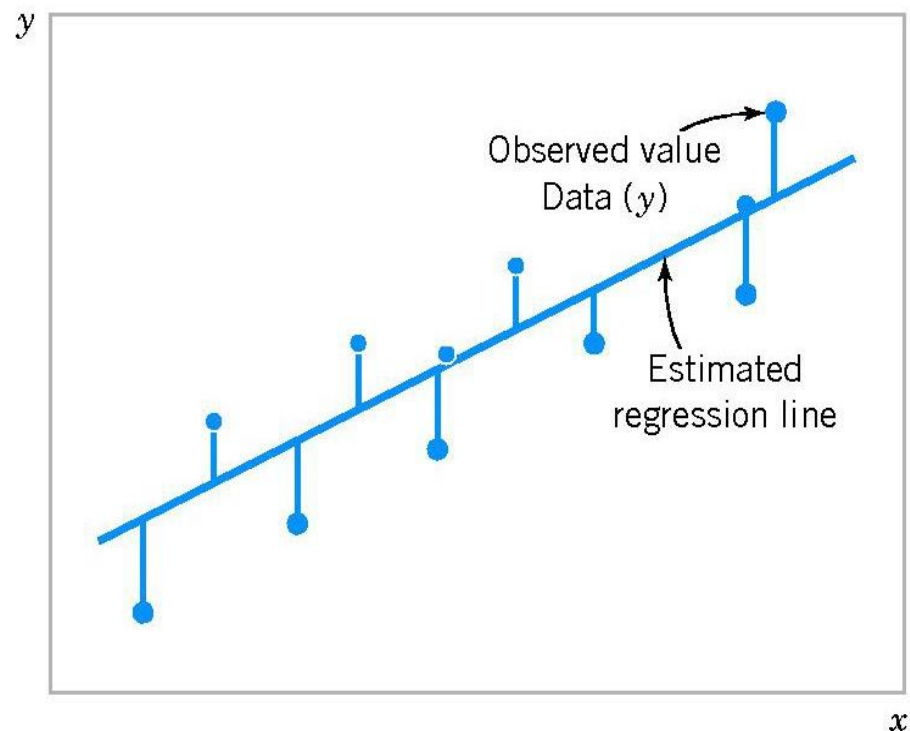
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The **method of least squares** is used to estimate the parameters, β_0 and β_1 by minimizing the sum of the squares of the vertical deviations in Figure 11-3.

Figure 11-3
Deviations of the data from the estimated regression model.



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The sum of the squares of the deviations of the observations from the true regression line is

$$L = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

The least squares estimators of β_0 and β_1 , say, $\hat{\beta}_0$ and $\hat{\beta}_1$, must satisfy

$$\left. \frac{\partial L}{\partial \beta_0} \right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\left. \frac{\partial L}{\partial \beta_1} \right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0$$

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Simplifying these two equations yields

$$\begin{aligned} n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_i &= \sum_{i=1}^n y_i \\ \hat{\beta}_0 \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n y_i x_i \end{aligned}$$

$\hat{\beta}_0, \hat{\beta}_1 = ?$

Notation

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i \right)^2}{n}$$

$$S_{xy} = \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^n x_i y_i - \frac{\left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n}$$

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Theorem

The **least squares estimates** of the intercept and slope in the simple linear regression model are

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

Estimated regression line is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

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Table 11-1 Oxygen and Hydrocarbon Levels

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6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
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We will fit a simple linear regression model to the oxygen purity data in Table 11-1. The following quantities may be computed:

$$n = 20 \quad \sum_{i=1}^{20} x_i = 23.92 \quad \sum_{i=1}^{20} y_i = 1,843.21 \quad \bar{x} = 1.1960 \quad \bar{y} = 92.1605$$

$$\sum_{i=1}^{20} y_i^2 = 170,044.5321 \quad \sum_{i=1}^{20} x_i^2 = 29.2892 \quad \sum_{i=1}^{20} x_i y_i = 2,214.6566$$

$$S_{xx} = \sum_{i=1}^{20} x_i^2 - \frac{\left(\sum_{i=1}^{20} x_i\right)^2}{20} = 29.2892 - \frac{(23.92)^2}{20} = 0.68088$$

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$$S_{xy} = \sum_{i=1}^{20} x_i y_i - \frac{\left(\sum_{i=1}^{20} x_i \right) \left(\sum_{i=1}^{20} y_i \right)}{20} = 2,214.6566 - \frac{(23.92)(1,843.21)}{20} = 10.17744$$

Therefore, the least squares estimates of the slope and intercept are

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{10.17744}{0.68088} = 14.94748$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 92.1605 - (14.94748)1.196 = 74.28331$$

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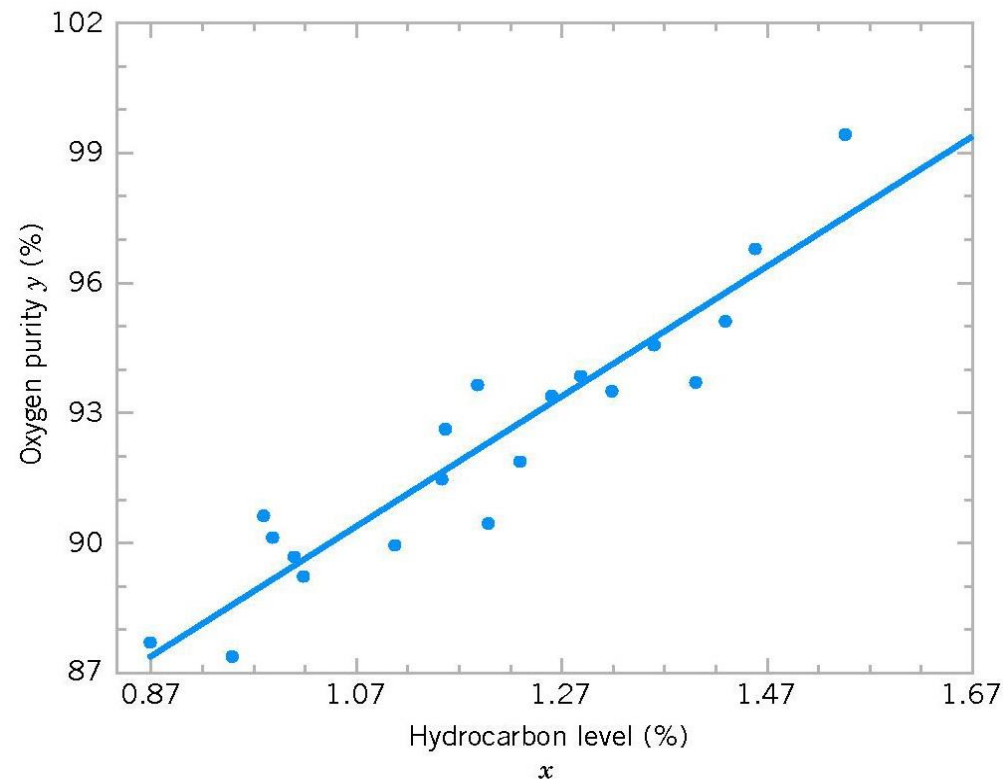
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The fitted simple linear regression model is

$$\hat{y} = 74.283 + 14.947x$$



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The error sum of squares is

$$SS_E = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

We have

$$E(SS_E) = (n - 2)\sigma^2.$$

$$SS_E = SS_T - \hat{\beta}_1 S_{xy}$$

$$SS_T = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n\bar{y}^2$$

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An **unbiased estimator** of σ^2 is

$$\hat{\sigma}^2 = \frac{SS_E}{n - 2}$$

$\hat{\sigma}$

Standard error

where

$$SS_E = SS_T - \hat{\beta}_1 S_{xy}$$

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Test on the β_1

$$H_0: \beta_1 = \beta_{1,0}$$

$$H_1: \beta_1 \neq \beta_{1,0}$$

Test statistic

$$T_0 = \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\hat{\sigma}^2 / S_{xx}}}$$

has the t distribution with $n - 2$ degrees of freedom.

If $|t_0| > t_{\alpha/2, n-2}$: reject H_0

If $|t_0| < t_{\alpha/2, n-2}$: fail to reject H_0

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Test on the β_1

An important special case

$$H_0: \beta_1 = 0$$

$$H_1: \beta_1 \neq 0$$

These hypotheses relate to the **significance of regression**.

Failure to reject H_0 is equivalent to concluding that there is no linear relationship between x and Y .

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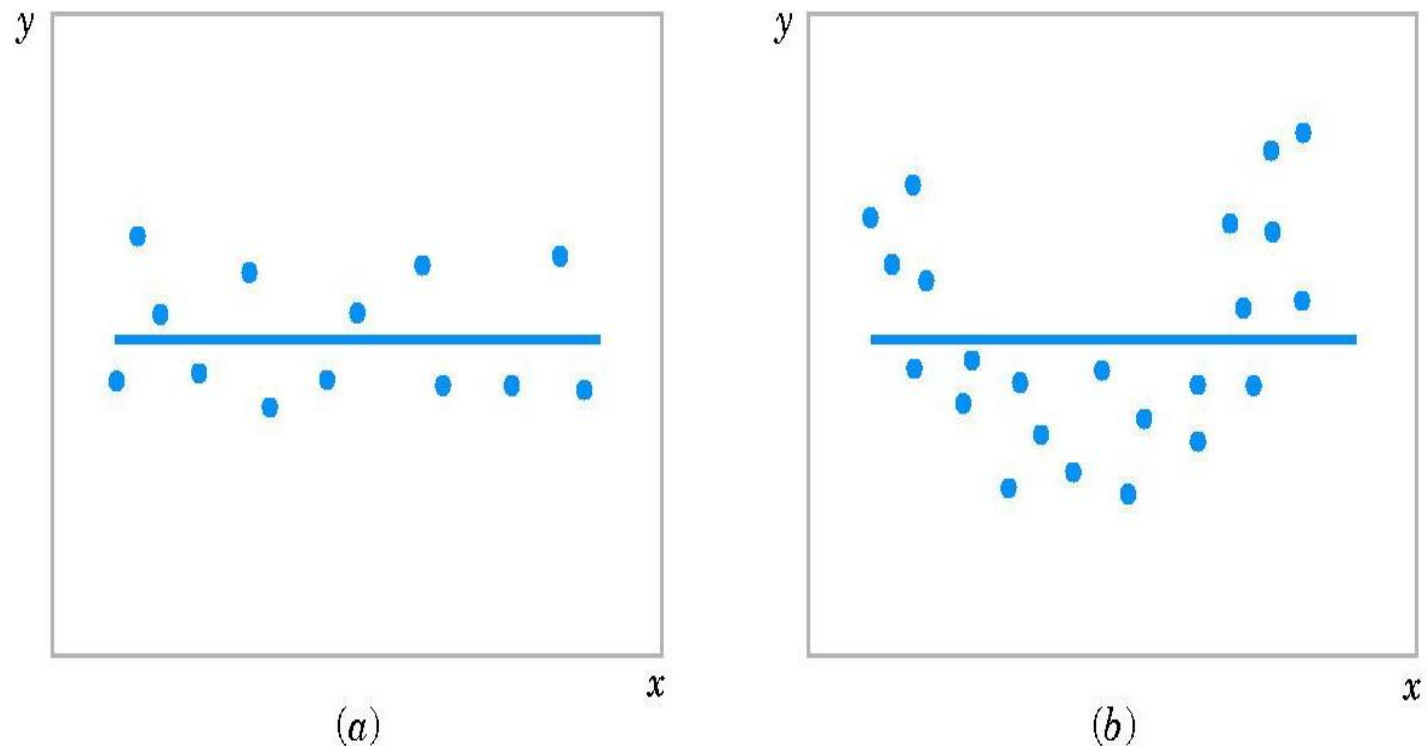


Figure 11-5 The hypothesis $H_0: \beta_1 = 0$ is not rejected.

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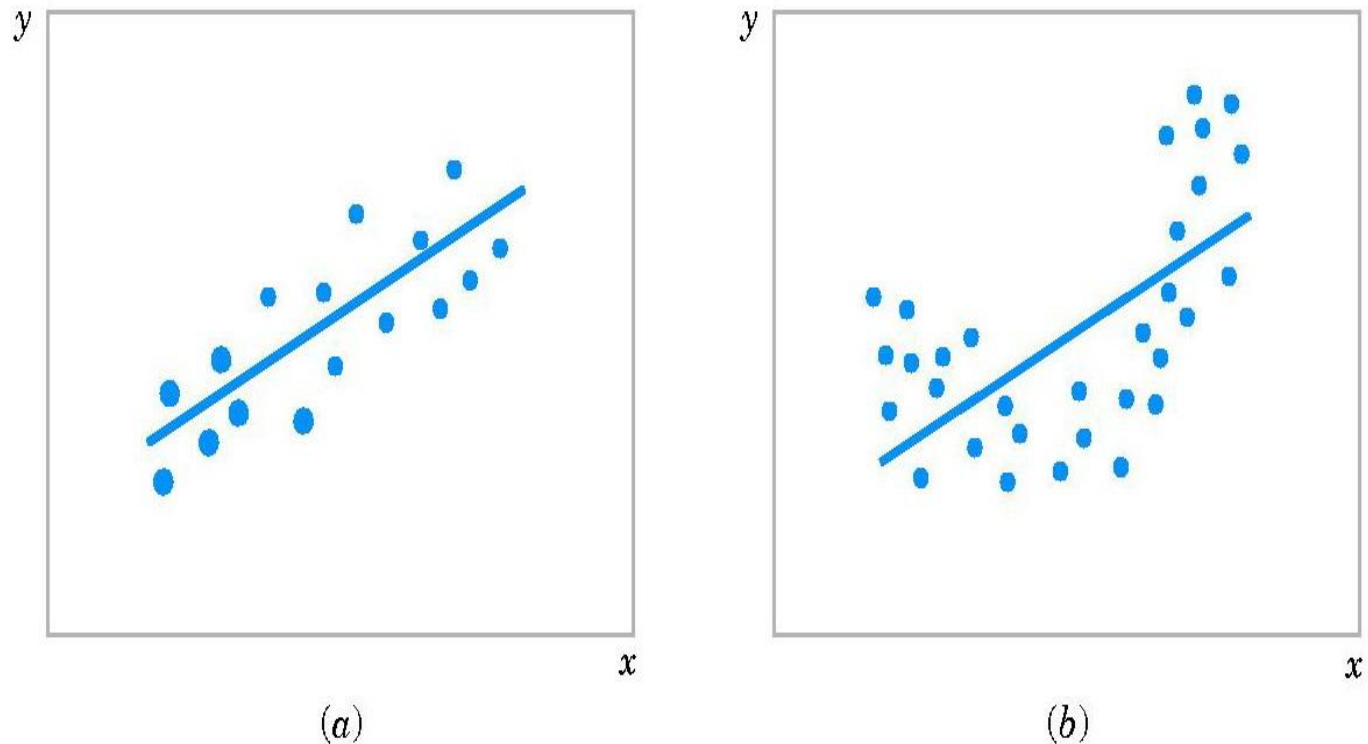


Figure 11-6 The hypothesis $H_0: \beta_1 = 0$ is rejected.

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We will test for significance of regression using the model for the oxygen purity data from Table 11-1. The hypotheses are

$$H_0: \beta_1 = 0$$

$$H_1: \beta_1 \neq 0$$

and we will use $\alpha = 0.01$.

Recall $\hat{\beta}_1 = 14.97$ $n = 20$, $S_{xx} = 0.68088$, $\hat{\sigma}^2 = 1.18$

Test statistic $t_0 = \frac{\hat{\beta}_1}{\sqrt{\hat{\sigma}^2/S_{xx}}} = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)} = \frac{14.947}{\sqrt{1.18/0.68088}} = 11.35$

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$$t_{\alpha/2, n-2} = t_{0.005, 18} = 2.88 < |t_0|$$



Reject H_0

If $|t_0| > t_{\alpha/2, n-2}$: reject H_0

If $|t_0| < t_{\alpha/2, n-2}$: fail to reject H_0

Test on the β_0

$$H_0: \beta_0 = \beta_{0,0}$$

$$H_1: \beta_0 \neq \beta_{0,0}$$

Test statistic

$$T_0 = \frac{\hat{\beta}_0 - \beta_{0,0}}{\sqrt{\hat{\sigma}^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right]}} = \frac{\hat{\beta}_0 - \beta_{0,0}}{se(\hat{\beta}_0)}$$

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Confidence Intervals on the Slope and Intercept

Under the assumption that the observations are normally and independently distributed, a $100(1-\alpha)\%$ confidence interval on the slope β_1 in simple linear regression is

$$\hat{\beta}_1 - t_{\alpha/2, n-2} \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \leq \beta_1 \leq \hat{\beta}_1 + t_{\alpha/2, n-2} \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

Similarly, a $100(1-\alpha)\%$ confidence interval on the intercept β_0 is

$$\hat{\beta}_0 - t_{\alpha/2, n-2} \sqrt{\hat{\sigma}^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right]} \leq \beta_0 \leq \hat{\beta}_0 + t_{\alpha/2, n-2} \sqrt{\hat{\sigma}^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right]}$$

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We will find a 95% confidence interval on the slope of the regression line using the data in Table 11-1.

Recall $\hat{\beta}_1 = 14.947$, $S_{xx} = 0.68088$, and $\hat{\sigma}^2 = 1.18$

CI 95% for β_1 :

$$\hat{\beta}_1 - t_{0.025,18} \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \leq \beta_1 \leq \hat{\beta}_1 + t_{0.025,18} \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

$$14.947 - 2.101 \sqrt{\frac{1.18}{0.68088}} \leq \beta_1 \leq 14.947 + 2.101 \sqrt{\frac{1.18}{0.68088}}$$

$$12.197 \leq \beta_1 \leq 17.697$$

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Confidence Interval on the Mean Response

$$\hat{\mu}_{Y|x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

A 100(1- α)% confidence interval about the mean response at the value of $x=x_0$ is given by

$$\hat{\mu}_{Y|x_0} - t_{\alpha/2, n-2} \sqrt{\hat{\sigma}^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]} \leq \mu_{Y|x_0} \leq \hat{\mu}_{Y|x_0} + t_{\alpha/2, n-2} \sqrt{\hat{\sigma}^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]}$$

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We will find a 95% confidence interval about the mean response for the data in Table 11-1.

The fitted model is $\hat{\mu}_{Y|x_0} = 74.283 + 14.947x_0$,

If we are interested in predicting mean oxygen purity when $x_0 = 100\%$ then

$$\hat{\mu}_{Y|x_{1.00}} = 74.283 + 14.947(1.00) = 89.23$$

CI 95% on $\mu_{Y|x_0}$

$$\hat{\mu}_{Y|x_0} \pm 2.101 \sqrt{1.18 \left[\frac{1}{20} + \frac{(x_0 - 1.1960)^2}{0.68088} \right]}$$

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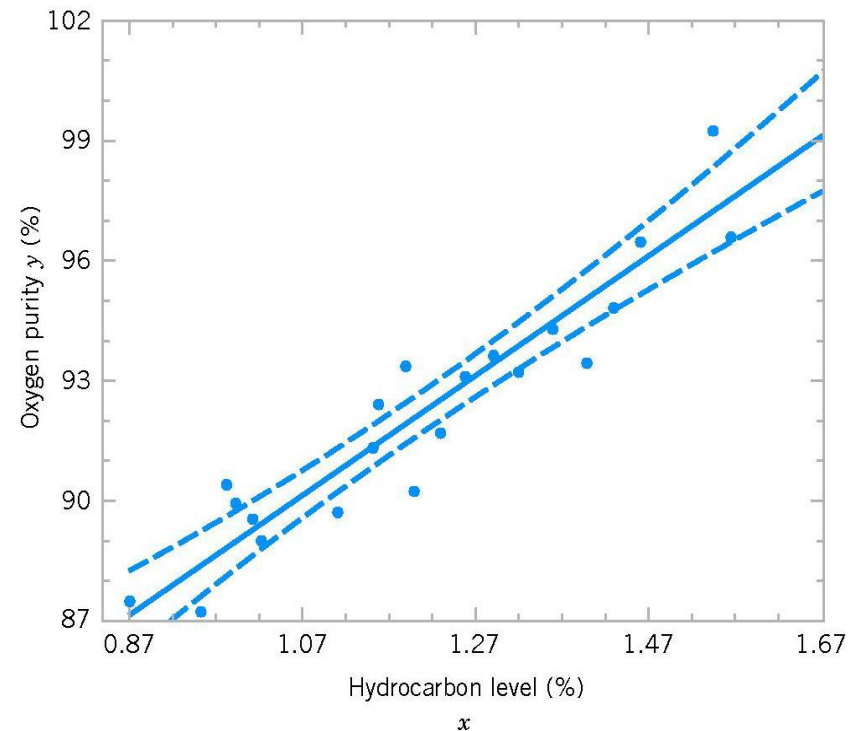
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$$\left\{ 89.23 \pm 2.101 \sqrt{1.18 \left[\frac{1}{20} + \frac{(1.00 - 1.1960)^2}{0.68088} \right]} \right\}$$

$$88.48 \leq \mu_{Y|1.00} \leq 89.98$$

Scatter diagram of oxygen purity with fitted regression line and 95% confidence limits on $\mu_{Y|x0}$.



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A $100(1-\alpha)\%$ prediction interval on a future observation Y_0 at the value x_0 is given by

$$\hat{y}_0 - t_{\alpha/2, n-2} \sqrt{\hat{\sigma}^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]} \leq Y_0 \leq \hat{y}_0 + t_{\alpha/2, n-2} \sqrt{\hat{\sigma}^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]}$$

Return to Table 11.1, confidence interval 95% on Y_0 at $x_0 = 100\%$

$$89.23 - 2.101 \sqrt{1.18 \left[1 + \frac{1}{20} + \frac{(1.00 - 1.1960)^2}{0.68088} \right]} \leq Y_0 \leq 89.23 + 2.101 \sqrt{1.18 \left[1 + \frac{1}{20} + \frac{(1.00 - 1.1960)^2}{0.68088} \right]}$$

$86.83 \leq y_0 \leq 91.63$

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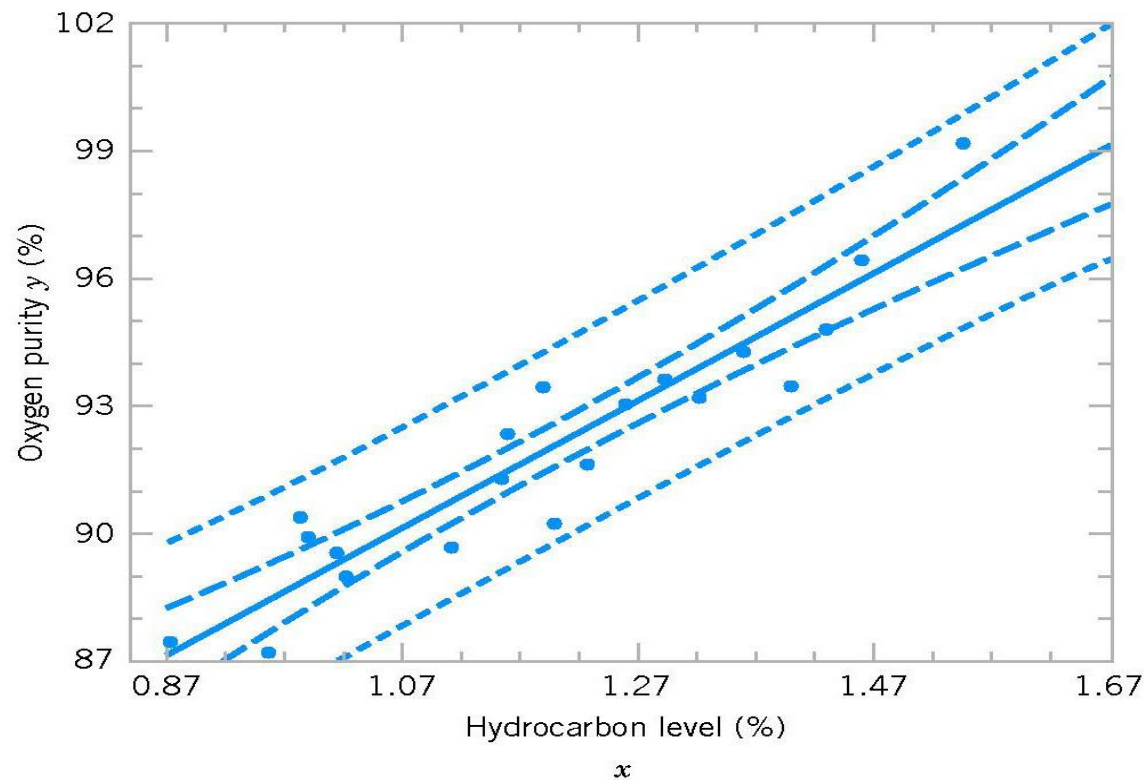
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Scatter diagram of oxygen purity data from Table 11.1 with fitted regression line, 95% prediction limits, and 95% confidence limits on $\mu_{Y|x0}$.

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- Fitting a regression model requires several **assumptions**.
 1. Errors are uncorrelated random variables with mean zero;
 2. Errors have constant variance; and,
 3. Errors be normally distributed.
- The analyst should always consider the validity of these assumptions to be doubtful and conduct analyses to examine the adequacy of the model

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Coefficient of Determination (R^2)

- The quantity

$$R^2 = \frac{SS_R}{SS_T} = 1 - \frac{SS_E}{SS_T}$$

is called the **coefficient of determination** and is often used to judge the adequacy of a regression model.

$$SS_R = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \hat{\beta}_1 S_{xy}.$$

- $0 \leq R^2 \leq 1$;
- We often refer to R^2 as the amount of variability in the data explained or accounted for by the regression model.

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For the oxygen purity regression model,

$$\begin{aligned} R^2 &= SS_R / SS_T \\ &= 152.13 / 173.38 \\ &= 0.877 \end{aligned}$$

Thus, the model accounts for 87.7% of the variability in the data.

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Definition

The **sample correlation coefficient**

$$R = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}} = \frac{S_{XY}}{\sqrt{S_{XX} SS_T}}$$

Note that

$$\hat{\beta}_1 = \left(\frac{SS_T}{S_{XX}} \right)^{1/2} R$$

We may also write:

$$R^2 = \hat{\beta}_1^2 \frac{S_{XX}}{S_{YY}} = \frac{\hat{\beta}_1 S_{XY}}{SS_T} = \frac{SS_R}{SS_T}$$

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Properties of the Linear Correlation Coefficient r

1. $-1 \leq r \leq 1$
2. The value of r does not change if all values of either variable are converted to a different scale.
3. The value of r is not affected by the choice of x and y . Interchange all x - and y -values and the value of r will not change.
4. r measures strength of a linear relationship.

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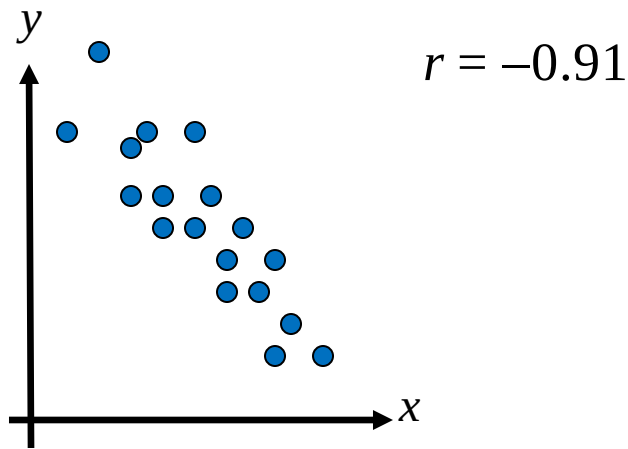
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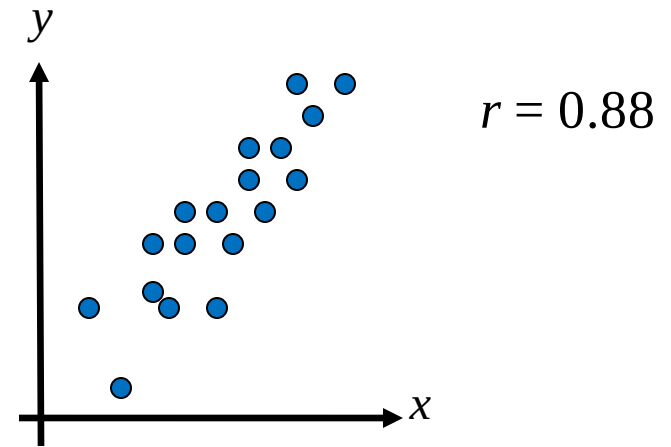
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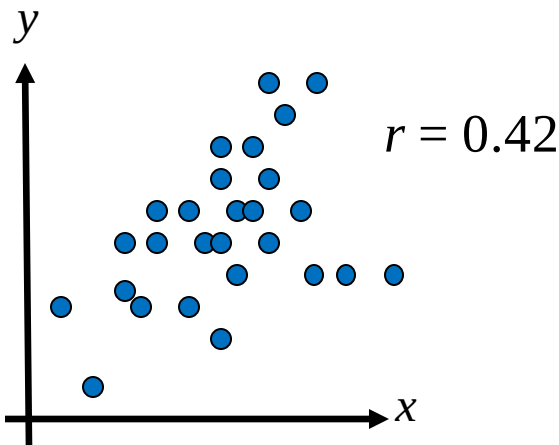
Summary



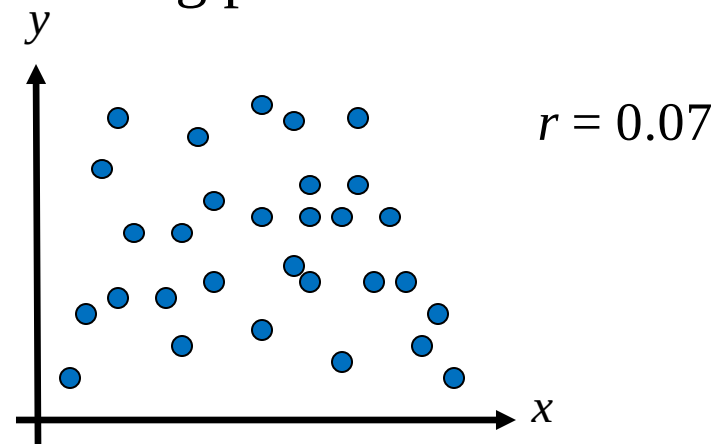
Strong negative correlation



Strong positive correlation



Weak positive correlation



Nonlinear Correlation

Empirical models

Simple Linear
Regression

Estimating σ^2
Hypothesis
tests

Confidence
intervals

Prediction
Adequacy

Correlation

Summary

Test on the ρ

$$H_0: \rho = 0$$

$$H_1: \rho \neq 0$$

Test statistic
$$T_0 = \frac{R\sqrt{n-2}}{\sqrt{1-R^2}}$$

has the t distribution with $n - 2$ degrees of freedom.

If $|t_0| > t_{\alpha/2, n-2} : \text{reject } H_0$

If $|t_0| < t_{\alpha/2, n-2} : \text{fail to reject } H_0$

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Summary

Test on the ρ

$$H_0: \rho = \rho_0$$

$$H_1: \rho \neq \rho_0$$

Test statistic $Z_0 = (\operatorname{arctanh} R - \operatorname{arctanh} \rho_0) \sqrt{n-3}$

$$\tanh u = (e^u - e^{-u}) / (e^u + e^{-u})$$

If $|t_0| > z_{\alpha/2}$: reject H_0

If $|t_0| < z_{\alpha/2}$: fail to reject H_0

Empirical models

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Summary

We have studied:

1. Empirical Models
2. Simple Linear Regression
3. Correlation

Homework: Read slides of the next lecture.